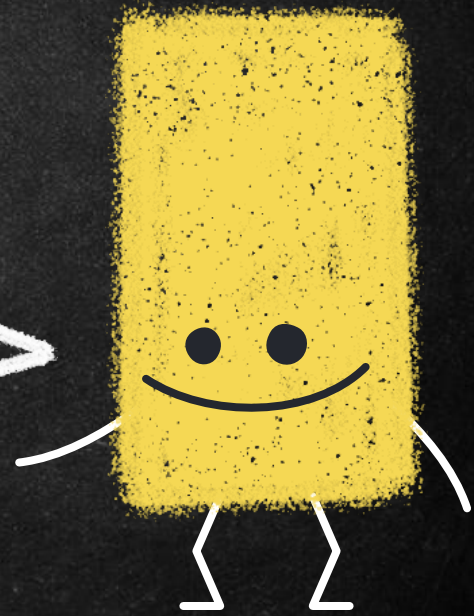
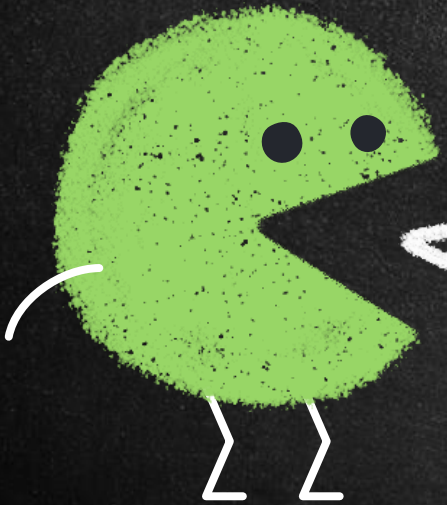


Digital Forensics

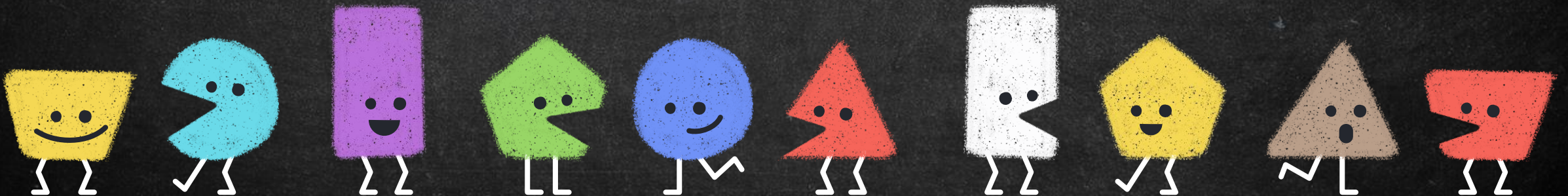
Unit – 2



Hello!

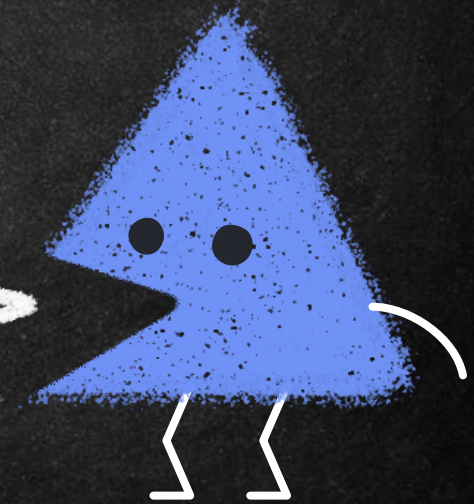
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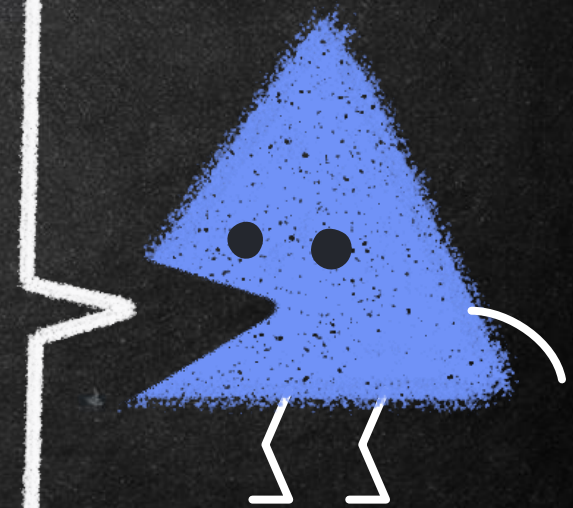
Chain of Custody

- A computer taken in as evidence makes many stops on its road to trial. It's collected, logged in at the lab, stored, checked out for analysis, checked back in for storage, and so on.
- Each of these stops must be noted, tracking each and every time the evidence item changes hands or locations.
- Without this detailed accounting, the evidence will be deemed untrustworthy and inadmissible.
- It's this detailed trail that makes up the chain of custody.



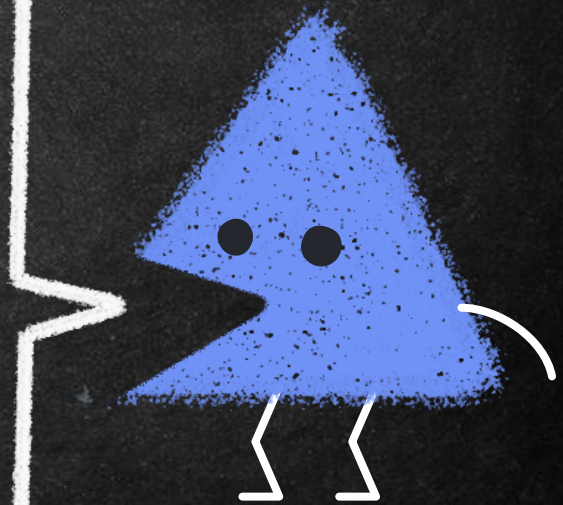
Hash Value

- hash is a unique value generated by a cryptographic hashing algorithm.
- Hash values (functions) are used in a variety of ways including cryptography and evidence integrity.
- Hash values are commonly referred to as a “digital fingerprint” or “digital DNA.”
- Any change to the hard drive, even by a single bit, will result in a radically different hash value. Therefore, any tampering or manipulation of the evidence is readily detectable.
- The most common hash functions used in digital forensics are Message Digest 5 (MD5), and Secure Hashing Algorithm (SHA) 1 and 2.



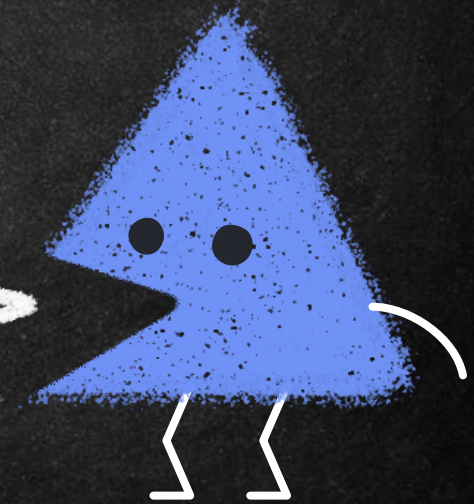
Use of Hash Value

- Hash values can be used throughout the digital forensic process.
- They can be used after the cloning process to verify that the clone is indeed an exact duplicate.
- They can also be used as an integrity check at any point it is needed.
- Examiners often have to exchange forensic images with the examiner on the opposing side. A hash value is sent along with the image so that it can be compared with the original.



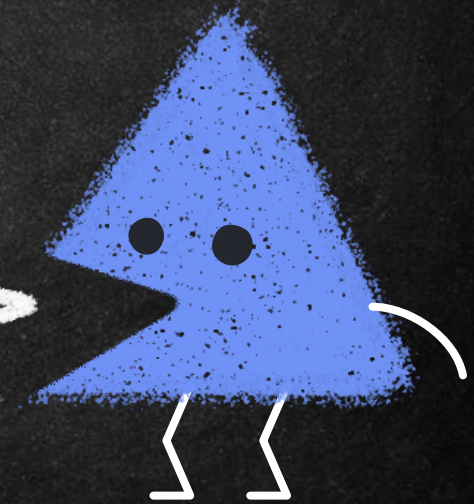
Use of Hash Value

- This comparison verifies that the image is a bit for bit copy of the original.
- The relevant hash values that were generated and recorded throughout the case should be kept and included with the final report. These digital fingerprints are crucial to demonstrating the integrity of the evidence and ultimately getting them before the jury.



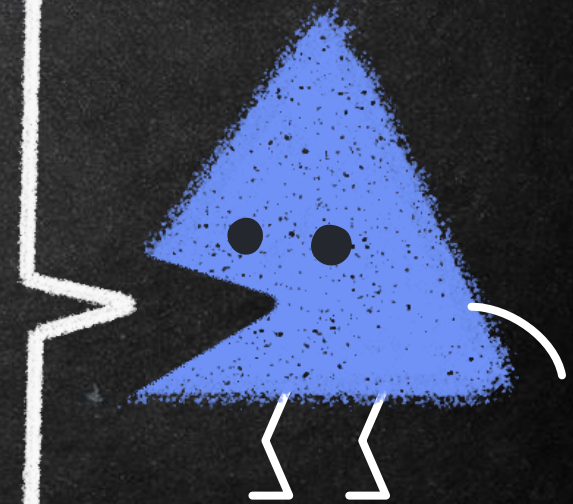
File Hashing

- File hashing is the act of attempting to uniquely identify a file.
- Different hash algorithms can be used for this purpose, but some are weaker than others and are more likely to have collisions.
- Once a file has been hashed, it can be compared with other file hashes.
- A common technique in incident response is to search for a file's hash value on Google and determine if the file's been seen before, and if so what its purpose is.



File Hashing

- Another technique used in contraband cases is to match a file's hash value with that of known contraband images to determine if the file had been seen before.
- In the case of trying to determine if a file is child sexually abusive material, identifying a file by hash value can help law enforcement prove the images are of a known victim.
- File hashing can also be used to whitelist files based on their signature. The white-listing process can result in data reduction in a case by removing legitimate files from a timeline. Doing so runs the risk of removing data from the analyst's view because attackers may use



References

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